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Fuse MetaFormer with convolutional neural networks for three-dimensional model classification

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ABSTRACT

Multimedia technology is widely applied to artificial intelligence and it is key to the performance of artificial intelligence systems, such as computer-aided design system, virtual reality system, the augmented reality system, medical image processing system, and game development system. With the development of multimedia technology, the number of three-dimensional (3D) models in network or database is becoming larger and larger. It is important to classify 3D models. In order to improve accuracy of three-dimensional model classification, a method of 3D model classification fusing MetaFormer with CNN (Convolutional Neural Networks) is proposed. 3D model is projected into two-dimensional (2D) views through the fixed-view projection, and representative views are selected by the clustering algorithm. Points sampled randomly from representative view are used to calculate its shape descriptors. View feature is extracted from representative view by MetaFormer. Shape feature is extracted from shape descriptors of representative view by Convolutional Neural Networks. View feature and shape feature of representative view are fused. At the same time, majority voting algorithm is used to determine category of 3D model based on the fusion of view features and shape features. Experiments are conducted on ModelNet10 dataset. Experimental results show that the proposed method achieves better results than others.

1. Introduction

With the development of 3D modeling technology and computer vision, the number and complexity of 3D models are growing exponentially. Now, it has become a hot research topic how to effectively classify and manage 3D models. In traditional methods, shape descriptors are defined manually, and then machine learning technology is adopted. However, there are some obvious problems. Designers should have rich experience to define shape descriptors manually. So, it lacks objectivity. At the same time, this method is difficult to capture deep features from 3D model. It mainly focuses on geometric properties of 3D model's surface, and ignores its internal structure and semantic information. In recent years, a series of CNNs have emerged in deep learning field, such as DenseNet (Huang et al., 2017), GoogleNet (Szegedy et al., 2015), ReMIT (Liu et al., 2025), etc. These CNNs have achieved remarkable results in tasks such as image processing and natural language processing. They can capture 3D model's characteristics better by fusing geometric information and semantic information to improve accuracy of 3D model classification. However, it faces huge challenges to apply CNNs directly to 3D model classification because of 3D model's high dimensionality, disorder and complex geometric structure. There are obvious differences between 3D model and 2D image, and traditional CNNs cannot directly deal with 3D models. MetaFormer, as general Transformer structure, is good at modeling relationships between features and aggregating features. MetaFormer is used to extract view features to describe 3D model's local details better. One dimensional CNN is more efficient for extracting shape features from shape descriptors D1, D2 and D3 to describe 3D model's shape and structure from different perspectives. When view features extracted by MetaFormer and shape features extracted by CNN are fused, 3D model can be described comprehensively. It not only makes up for the shortcomings of view features and shape features, but also maintains high robustness under shape transformation and perspective change, which improves accuracy of 3D model classification effectively. Highlights of this paper are shown as follows:

(1) Mini Batch K-Means algorithm is adopted to extract representative views from 24 projection views of 3D model for reducing redundant information.

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(2) MetaFormer is used to extract view features from representative views to describe local details of 3D model better from different perspectives. CNN is used to extract shape features from shape descriptors D1, D2, D3 to express shape and structure of 3D model from different perspectives. They are fused to provide comprehensive information about 3D model.
(3) Majority voting algorithm is adopted to determine category of 3D model based on the fusion of view features and shape features.

2. Related work

3D model representation is very complex. Currently, there are three main solutions for 3D model classification, including voxel-based methods, point cloud-based ones and view-based ones.

In voxel-based methods, 3D model is denoted as the voxel matrix, and 3D Convolutional Neural Network (3D CNN) is used to extract deep features from voxel matrix. It can enhance effectively the ability of feature extraction, which significantly improves accuracy of 3D model classification. Yang et al. (2019) apply the voxelization technology to convert 3D polygon grid into voxel matrix, from which deep voxel CNN is used to extract discriminative features. Cai et al. (2021) propose a parallel network based on voxel and 3D views, which performs the weighted fusion of voxels and views. Kim et al. (2021) design Part Geometry Network (PG-Net), which is used as feature descriptor for 3D model classification and reconstruction. Ma et al. (2022) propose dual-channel attention residual network, which uses voxels and 3D Radon feature matrix to describe 3D model's global and local information. 3D CNNs can extract local information directly from the voxelized data, but the voxel-based classification methods are more expensive when 3D CNNs are used to extract features. At the same time, accuracy of 3D model classification is affected by the voxel resolution.

In point cloud-based methods, point cloud is preprocessed to solve its sparseness and disorder, and then neural network is used to extract discriminative features for classifying 3D model. Guo et al. (2021) design the fusion network to classify and segment point clouds, which includes global feature extractor, local feature extractor and adaptive feature mixer. Liang et al. (2023) use PointNet++ to build deep learning model to improve the stability and robustness of point cloud classification. Gao et al. (2021) propose a new method of 3D model classification based on multi-head self-attention, which consumes sparse point clouds and learns their feature representations. Hassan et al. (2023) introduce deep hierarchical architecture of point cloud classification, which consists of multi-layer sampling, concentric ring convolution, pooling layer and residual feature propagation block. Song et al. (2022) propose Kernel Correlation Learning Block (KCB), which can adaptively learn local and global features at different levels. Point clouds can be used to describe position, size and geometric information of 3D model comprehensively. However, point cloud-based methods are affected by its sparseness and disorder.

In view-based classification methods, a set of 2D views are used to describe 3D model. 2D view is simple expression of 3D model, and it is relatively simple to extract features from 2D views. View features are extracted from 2D views by neural networks, and features from different perspectives are merged by pooling operations to classify 3D models. Jin and Li (2022) propose the framework of 3D model representative view position based on the prediction of rotation direction and rotation angle. Guan et al. (2023) design the network jointly driven by multi-modal features and word embeddings, which is used to represent, classify and retrieve 3D model. Liang et al. (2020) use Multi-View Convolutional LSTM Network (MVCLN) to extract view's temporal and spatial information. Bai et al. (2019) apply CaffeNet to 2D view classification and adopts the weighted voting algorithm for 3D model classification. Zhang et al. (2018) propose inductive multi-hypergraph learning algorithm to obtain optimal projections from 3D model. Huang et al. (2020) present a view-based weight network for 3D object recognition, where the view-based weights are assigned to different projections. Zhu et al. (2022) propose local information fusion network to use the relationship between image areas and feature mapping channels.

Transformer has shown great potential in computer vision tasks, and it is generally believed that token mixer has made the greatest contribution to Transformer (Vaswani et al., 2017). However, recent work shows that attention-based modules in Transformers can be replaced by spatial MLP, and its performance is good (Tolstikhin et al., 2021). Yu et al. (2022) give a general architecture called as MetaFormer from Transformers and demonstrates that it can improve classification accuracy.

In view-based classification methods, a large amount of redundant information appears when 3D model is projected into 2D views. At the same time, 2D view only represents partial information on 3D model's surface and can't accurately denote its spatial structure. This paper proposes a method of 3D model classification which combines MetaFormer and CNN. Firstly, 3D model is projected into a series of 2D views, from which Mini Batch K-means algorithm is used to extract representative views. MetaFormer is used to extract view features from representative views. View features are local and do not emphasize contour boundaries. Points are randomly sampled from the outline boundary. Statistical methods are used to calculate shape descriptors D1, D2, and D3 of representative view, from which CNN is adopted to extract shape features. Shape features are global, which emphasizes contour boundaries and makes up for the shortcomings of view features (Osada et al., 2001). View features extracted by MetaFormer and shape features extracted by CNN are fused to describe 3D model better. Finally, softmax function is adopted to determine category of representative view, and majority voting algorithm is used to classify 3D model. Its main contributions are shown as follows:

(1) Mini Batch K-means algorithm is used to extract representative views from 2D views of 3D model.
(2) MetaFormer is adopted to extract view features from representative view, and CNN is used to extract shape features from shape descriptors D1, D2, and D3 of representative view.
(3) View features and shape features are combined to represent 3D model, and then Softmax function and majority voting algorithm are used to determine category of 3D model.

The novelty of the proposed method is that clustering algorithm is used to extract representative views for reducing redundant information, view features and shape features are extracted from representative views, and majority voting algorithm is applied to classify 3D models.

3. Representation of 3D model

3.1. 2D representative view

3D models are complex. It is difficult to express and process 3D models. In view-based classification methods, 3D model is usually projected into a series of 2D views. This article adopts the fixed projection method based on a sphere. 3D model is fixed at the center of the sphere, and virtual camera is placed on 60°circumference obliquely above 3D model. 3D model is taken photos along the circumference to obtain the projection views from different perspectives. 3D model rotates one circle at 15°per step, and 24 views are obtained. The fixed projection method can render 3D model and obtain its shape information comprehensively. For model 'bed', the fixed projection method is used to extract its 2D views, as shown in Fig. 1.

A large number of 2D views are generated at positions with uniform distribution around 3D model. It results in serious accumulation of redundant information and thus reduces accuracy of 3D model classification. At the same time, if all views are treated equally, the difference of importance between views will be ignored.

In this paper, Mini Batch K-means algorithm is adopted to cluster 2D views for selecting representative views. Representative view is selected

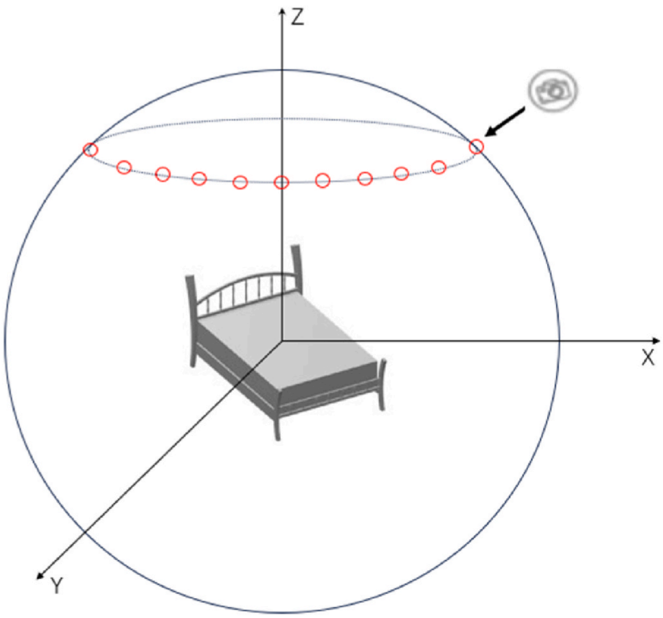

Fig. 1. 2D projection of 3D model.

from each cluster to obtain a set of representative views. This method improves the ability of 3D model representation and reduces the number of 2D views. So, accuracy of 3D model classification is improved.

3.2. Shape descriptior D1, D2, D3

Osada et al. (2001) use shape distribution to describe 3D model. This paper gives shape descriptor suitable for 2D view based on 3D shape distribution. Points are randomly sampled from the outline of representative view. Although more points can fully represent 3D model's shape, computation cost is too big. Here, the fixed number of sampling points are used, and the number of sampling points is 1024. Fig. 2 shows shape descriptors D1, D2 and D3 for representative view respectively.

N points are sampled from the outline of representative view to form PD1 = $\{m_{i1}, \ldots, m_{ik}, \ldots, m_{iN}\}$, and D1 shape descriptor is $\{D1_1, \ldots, D1_i, \ldots, D1_{Bins}\}$. $D1_i$ is calculated as shown in formula (1).

$$D1_i = |\{m|dist(m, O) \in (BSize \times (i-1), BSize \times i), m \in PD1\}| \tag{1}$$

where, BSize = max ({dist (m, O)|m ∈ PD1})/Bins, dist () represents Euclidean distance between two points, and O is the centroid of view.

N point pairs are sampled from the outline of representative view to form PD2 = $\{(m_{i1}, n_{i1}), \ldots, (m_{ik}, n_{ik}), \ldots, (m_{iN}, n_{iN})\}$, and D2 shape descriptor is $\{D2_1, \ldots, D2_i, \ldots, D2_{Bins}\}$. $D2_i$ is calculated as shown in formula (2).

$$D2_i = |\{(m_{ik}, n_{ik})|dist(m_{ik}, n_{ik}) \in (BSize \times (i-1), BSize \times i), (m_{ik}, n_{ik}) \in PD2\}| \tag{2}$$

where, BSize = max ({dist (m_{ik}, n_{ik})|(m_{ik}, n_{ik}) ∈ PD2})/Bins.

N point triples are sampled from the outline of representative view to form PD3 = $\{(m_{i1}, n_{i1}, q_{i1}), \ldots, (m_{ik}, n_{ik}, q_{ik}), \ldots, (m_{iN}, n_{iN}, q_{iN})\}$, and D3 shape descriptor is $\{D3_1, \ldots, D3_i, \ldots, D3_{Bins}\}$. $D3_i$ is calculated as shown in formula (3).

$$\begin{aligned} D3_i = |\{(m_{ik}, n_{ik}, q_{ik})|herson(m_{ik}, n_{ik}, q_{ik}) \in (BSize \times (i-1), BSize \times i), \\ (m_{ik}, n_{ik}, q_{ik}) \in PD3\}| \end{aligned} \tag{3}$$

Where, BSize = max ({herson (m_{ik}, n_{ik}, q_{ik})|(m_{ik}, n_{ik}, q_{ik}) ∈ PD3})/Bins, and herson (m_{ik}, n_{ik}, q_{ik}) is calculated as shown in formula (4) and formula (5).

$$herson((m_{ik}, n_{ik}, q_{ik})) = \sqrt{s \times (s - l_1) \times (s - l_2) \times (s - l_3)} \tag{4}$$

$$s = \frac{l_1 + l_2 + l_3}{2} \tag{5}$$

Where, l_1 = dist (m_{ik}, n_{ik}), l_2 = dist (m_{ik}, q_{ik}), l_3 = dist (n_{ik}, q_{ik}).

D1, D2, D3 shape descriptors are connected in series to form $(D1_1, \ldots, D1_i, \ldots, D1_{Bins}, D2_1, \ldots, D2_i, \ldots, D2_{Bins}, D3_1, \ldots, D3_i, \ldots, D3_{Bins})$.

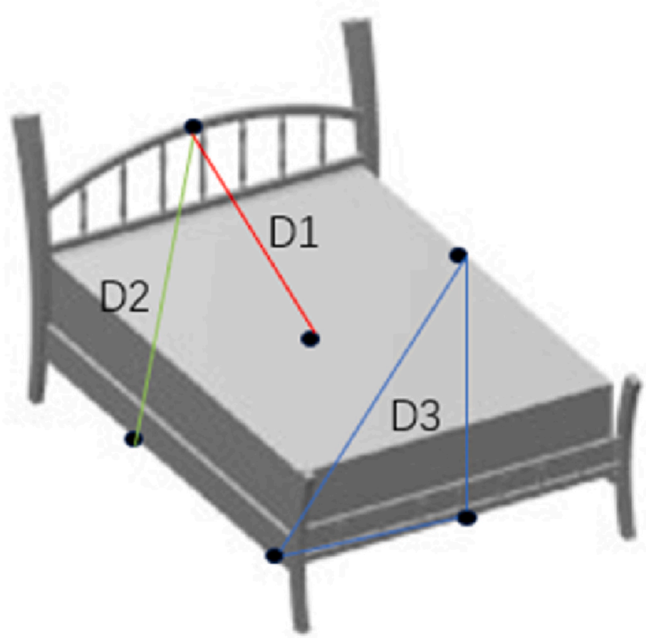

Fig. 2. Shape descriptors for representative view.

4. 3D model classification based on MetaFormer and CNN

The framework of 3D model classification in this paper is shown in Fig. 3, including data preprocessing, MetaFormer, CNN, flatten layer, merging layer, fully connected layer, Softmax function and voting algorithm.

3D model M is projected into $V_1, V_2, \ldots, V_{24}$ to obtain view set V(M) = $\{V_i|1 \le i \le 24\}$. V(M) is clustered to obtain representative view set VR (M) = $\{\dot{V}_i|1 \le i \le K\}$. Points are randomly sampled from the outline of representative views in VR(M), and shape descriptors D1, D2 and D3 are calculated. Representative views are input into MetaFormer to extract view features. Shape descriptors D1, D2, and D3 of representative views are input into CNN to extract shape features. Then, view features and shape features are processed by the flatten layer and then fused in the merging layer. Shape features are used to enhance the description ability of view features. Finally, the fused features are input into the fully connected layer, and Softmax function is used to classify representative views, based on which majority voting algorithm is used to determine category of M.

Mini Batch K-Means algorithm is a typical partition-based clustering algorithm. Euclidean distance is used as indicator to measure the similarity between 2D views, which is inversely proportional to the distance between views. The greater is the similarity, the smaller is the distance between views. Set initial cluster number K and K initial cluster centers, and divide V(M) into K clusters according to the distance between views. Cluster centers are updated continuously to reduce their sums of squared error (SSE). When SSE no longer changes or converges, the clustering process stops and clustering results are obtained. Mini Batch K-Means algorithm is an optimization variant of K-Means algorithm. The size of ModelNet10 and ModelNet40 dataset is very large. Mini Batch K-Means

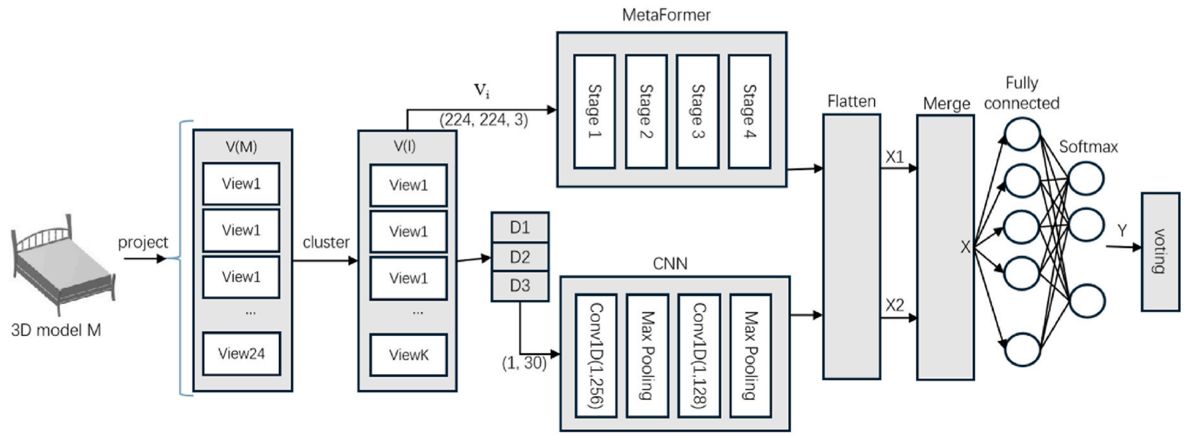

Fig. 3. 3D model classification based on MetaFormer and CNN.

algorithm reduces computation cost of each iteration by extracting small batches of data, which makes it more efficient on a large dataset. At the same time, memory usage is reduced. Compared with K-Means algorithm, Mini Batch K-Means algorithm converges faster, reduces the consumption of computation resources and maintains good clustering results.

The process of extracting representative views of 3D model M is shown as follows:

Step 1. Project M into 24 views, and collect them into $V(M) = \{V_i|1 \le i \le 24\}$;

Step 2. Select K views randomly from V(M) as initial center C_i, $U_i = \Phi$, $(1 \le i \le K)$, U=V(M);

Step 3. For(i = 1; i ≤ K; i++)

For (j = 1; j ≤ 24; j++)

$$d(V_j, C_i) = \sqrt{\sum_{k=1}^{m} (V_{jk} - C_{ik})^2} \tag{6}$$

where, V_{jk} and C_{ik} are the kth pixel values of views V_j and C_i respectively, and m is the number of pixels of V_j.

Step 4. For(i = 1; i ≤ K; i++)

$V = \underset{u \in U}{\mathrm{argmin}}\, d(u, C_i)$ is set into U_i, $U = U - \{V\}$;

Step 5. $C_i \leftarrow \frac{1}{|U_i|} \sum_{V \in U_i} V$, $(1 \le i \le K)$;

Step 6. Repeat from step 3 to step 5 continuously until SSE no longer changes or converges, and SSE is calculated as shown in formula (7).

$$SSE = \sum_{i=1}^{K} \sum_{V \in U_i} |d(V, C_i)|^2 \tag{7}$$

Step 7. Output $VR(M) = \{C_i|1 \le i \le K\}$.

MetaFormer is a general architecture abstracted from Transformer, and its structure is similar with Transformer. Researches on ImageNet dataset show that the performance of MetaFormer is significantly better than that of self-attention. MetaFormer is divided into four stages. Each stage consists of a Patch embedding layer and several MetaFormer Blocks. MetaFormer Block contains token mixer, such as IdentityMapping, RandomMixxing, Convolutions and Attention. MetaFormer structure is shown in Fig. 4, in which C1, C2, C3, and C4 represent the number of channels, and L represents the number of blocks.

Representative view V_i in VR(M) is processed by Patch operation (Dosovitskiy et al., 2021), as shown in formula (8):

$$M = InputEmb(V_i) \tag{8}$$

Where, $M \in R^{N \times C}$ represents the embedding of the input, N is the number of patch of V_i, and C represents the dimension of the embedding.

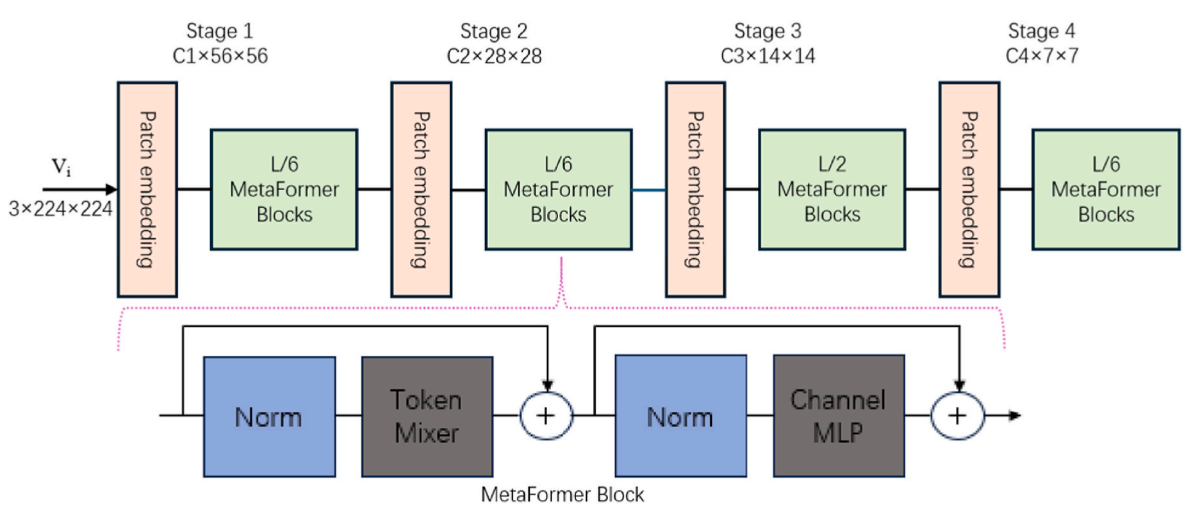

Fig. 4. MetaFormer structure.

In the first stage, the size of convolution kernel is 7 and the length of stride is 4. In other 3 stages, the size of convolution kernel is 3 and the length of stride is 2.

M is input into MetaFormer Block, which contains two residual sub-blocks. The first sub-block contains token mixer, in which information is changed between tokens, as shown in formula (9):

$$N = TokenMixer(Norm(M)) + M \tag{9}$$

Where, Norm () represents the normalization operation, and TokenMixer () is token mixer which is used to mix token information. It is implemented through various attention mechanisms in Transformer models or spatial MLP (Tolstikhin et al., 2021; Touvron et al., 2023).

There are several kinds of token mixers:

Identity mapping can be used as token mixer. Identity mapping does not actually mix tokens, as shown in formula (10):

$$IdentityMapping(X) = X \tag{10}$$

Random mixing can be used as token mixer, as shown in formula (11):

$$RandomMixing(X) = XW_R \tag{11}$$

Where, $W_R \in R^{N\times N}$ represents the frozen matrix after random initialization.

The reverse separable convolution in MobileNetV2 (Sandler et al., 2018) can be used as token mixer, as shown in formula (12):

$$Convolutions(X) = \mathrm{Convpw2}(\mathrm{Convdw}(\sigma(\mathrm{Convpw1}(X)))) \tag{12}$$

Where, Convpw1 () and Convpw2 () are point-wise convolutions, Convdw () is the deep convolution, σ() represents nonlinear activation function, the size of convolution kernel is 7, and the expansion ratio is 2.

Self-attention (Vaswani et al., 2017) in Transformer can also be used as token mixer.

The second sub-block consists of two-layer Channel MLP with nonlinear activation, as shown in formula (13):

$$Z = \sigma(\mathrm{Norm}(N)W_1)W_2 + N \tag{13}$$

Where, $W_1 \in R^{C\times rC}$ and $W_2 \in R^{rC\times C}$ are the learnable parameters of expansion ratio *r* in MLP. *C* is the number of channels, which is the number of the input features in the first layer MLP. *rC* is the number of input features in the second layer MLP, and the value is the product of *C* and *r*. σ() is nonlinear activation function, and it is ReLU activation function.

Shape descriptors D1, D2, and D3 are one dimensional. One dimensional convolutional neural network is more efficient for extracting shape feature from shape descriptors D1, D2 and D3. At the same time, when CNN is adopted, the proposed network can be trained faster, and better results of 3D model classification can be gotten. The relationship between locations of global shape feature can be viewed as one dimensional sequence. This paper uses one dimensional CNN to process global shape features. Two convolutional layers and two max-pooling layers are adopted. Two convolutional layers are Conv1D (1, 256) and Conv1D (1, 128). The size of convolution kernel is 1 and the length of stride is 1. ReLU activation function is adopted. Shape descriptors D1, D2, and D3 of representative views in V(M) are normalized. Each convolutional layer is followed by max pooling layer for down-sampling, which reduces computation complexity and parameter volume. At the same time, important features are retained. The input size of CNN is 1*30 and its output size is 1*128.

View features extracted from representative views by MetaFormer are processed by the flatten layer to obtain X_1. Shape features extracted by CNN from shape descriptors D1, D2, and D3 of representative views are processed by the flatten layer to obtain X_2. X_1 and X_2 are fused in the merging layer to obtain feature vector X, as shown in formula (14):

$$X = Merge(X_1, X_2) \tag{14}$$

Feature vector X is input into Softmax function, and probability distribution of view V_i under categories $s_1, s_2, \ldots, s_m$ are computed as shown in formula (15):

$$Y = SoftMax(WX + B) \tag{15}$$

Where, W and B are parameters in Softmax function, $Y = (P(s_1|V_i), P(s_2|V_i), \ldots, P(s_m|V_i))$.

Y is converted into one-hot encode R, as shown in formula (16):

$$R = one-hot(Y) \tag{16}$$

Views in $VR(M) = \{V_i^{'}|1 \le i \le K\}$ are input into the proposed network, and the output of Softmax function is converted into one-hot encode to obtain $R_1, R_2, \ldots, R_K$. Use $R_1, R_2, \ldots, R_K$ to construct matrix $T = \{R_{mn}| 1 \le m \le K, 1 \le n \le 10\}$, as shown in formula (17):

$$T = \begin{bmatrix} R_{11} & R_{12} & \ldots & R_{1,10} \\ R_{21} & R_{22} & \ldots & R_{2,10} \\ \vdots & \vdots & \vdots & \vdots \\ R_{K1} & R_{K2} & \ldots & R_{K,10} \end{bmatrix} \tag{17}$$

Where, R_{ij} indicates whether representative view $V_i^{'}$ belongs to category s_j. 1 means that $V_i^{'}$ belongs to category s_j, and 0 means that $V_i^{'}$ does not belong to category s_j.

The category of M is determined, as shown in formula (18):

$$Class(M) = \underset{n=1,2,\ldots,10}{\mathrm{argmax}} \sum_{m=1}^{K} R_{mn} \tag{18}$$

View features can describe 3D model's global shape information, but they are not able to express its contour details comprehensively. On the contrary, shape features can describe 3D model's local contour boundary better, but they can't represent it on the whole. When view features and shape features are fused, they can complement each other's strengths and weaknesses to better describe global shape and local contour details of 3D model. Then, the fused feature is input into Softmax function to compute probability distribution that view belongs to categories $s_1, s_2, \ldots, s_m$, based on which majority voting algorithm can determine category of 3D model more correctly.

5. Experimental results and analysis

ModelNet10 dataset is selected in experiments, which contains 10 categories including bathtub, bed, chair, desk, dresser, monitor, night_stand, sofa, table, toilet. ModelNet10 dataset includes 3991 training models and 908 test ones. 454 models are selected randomly from 3991 training models as validation set, and the rest 3537 training models are collected as training set. 908 test ones are collected as test set. In order to ensure the fairness and reliability of experiments, hyperparameters are optimized on validation set. Test set is only adopted for evaluating the proposed network and does not participate in parameter optimization. Six groups of experiments are conducted.

The first set of experiments study the effect of cluster number K on accuracy of the proposed network. 3D model is projected to obtain 24 views, and Mini Batch K-Means algorithm is used to obtain K representative views. MetaFormer is used to extract view features from representative views and representative views are classified. Finally, majority voting algorithm is used to determine the category of 3D model. When cluster number K is set to 4, 8, 12, 16, 20 and 24 respectively, accuracies, FLOPs and Params of the proposed network on validation set are shown in Table 1.

From Table 1, we can see that with cluster number K increasing, accuracy of the proposed network on validation set increases firstly and then decreases, and its FLOPs and Params increase. When K is 4, accuracy, FLOPs and Params of the proposed network are the lowest, which are respectively 87.2 %, 4.2G and 18.3M. When K is 20, its accuracy is the highest and the value is 88.5. When K is 24, its FLOPs and Params are the highest, which are respectively 17.1G and 37.0. Experimental results

Table 1
Accuracies, FLOPs and Params of the proposed network on validation set under different K value.

K	Accuracy	FLOPs(G)	Params (M)
4	0.872	4.2	18.3
8	0.879	5.8	21.7
12	0.883	8.7	25.6
16	0.881	11.5	29.4
20	0.885	14.3	33.2
24	0.880	17.1	37.0

on validation set show that moderate increase of K can retain more discriminative information of 2D views. When K is set to 24, Mini Batch K-Means algorithm is not performed and 24 views are all selected. Accuracy of the proposed network is 88.0 %. When K is set respectively to 12, 16, 20, accuracy of the proposed network is almost the same. But, with K value increasing, FLOPs and Params of the proposed network increase rapidly. When K is set to 12, accuracy, FLOPs and Params of the proposed network are respectively 88.3 %, 8.7G and 25.6M. Its accuracy decreases slightly, and its FLOPs and Params decrease greatly. So, K is set to 12 and 12 representative views are selected for optimizing parameters of the proposed network.

We use Mini Batch K-Means and K-Means respectively to select representative views. Then, comparative experiments are conducted on ModelNet10 set. The performance of K-Means and Mini Batch K-Means is shown in Table 2.

Different networks can be obtained by setting MetaFormer's token mixer. In the second group of experiments, when MetaFormer's token mixers are respectively set to IdentityMapping, RandomMixxing, Convolutions and Attention, and four derived networks are obtained. Each network includes four stages. 3D model is projected into 24 views, and Mini Batch K-Means algorithm is used to select 12 representative views. MetaFormer is used to extract view features from representative views and representative views are classified, based on which majority voting algorithm is used to determine category of 3D model. Accuracies of four derived networks on validation set are shown in Table 3.

From Table 3, we can see that when token mixer is depthwise separable convolutions, accuracy of the proposed network on validation set is the highest, reaching 91.2 %. When token mixer employs depthwise separable convolution, it separates spatial convolution from channel convolution. Separable convolution is used to process spatial features of each channel independently, and cross-channel information is fused by point-by-point convolution. This separation mechanism significantly reduces computation complexity, and retains richer local details and inter-channel correlations. Ordinary convolution considers the region firstly and then the channel. But, depthwise separable convolution achieves the separation of channels and regions, and can consider the changes of channels and regions at the same time. Depthwise separable convolution balances between the ability of capturing local geometric structures and the ability of modeling the dependency relationship of views. Depthwise separable convolution can better capture spatial relationship and correlation between views. In contrast, IdentityMapping lacks the ability of feature transformation, RandomMixxing relies on the fixed weights which are difficult to be optimized, and self attention is prone to the overfitting problem. So, token mixer of

Table 3
Accuracies of four derived networks under different token mixer on validation set.

Token mixer	Block number	Accuracy
IdentityMapping (Id,Id,Id,Id)	L=(1,1,3,1)	0.879
RandomMixxing (Id,Id,Rand, Rand)	L=(1,1,3,1)	0.887
Convolutions (Conv, Conv,Conv, Conv)	L=(1,1,3,1)	0.912
Attention (Conv, Conv,Attn, Attn)	L=(1,1,3,1)	0.905

L represents the number of blocks at different stages, Id is identity mapping, Rand represents random mixing, Conv is separable convolution, and Attn represents self-attention.

MetaFormer is set to depthwise separable convolutions in experiments.

The third group of experiments investigate the influence of the number of MetaFormer Blocks on the proposed network. The ratio of the number of MetaFormer Blocks contained in 4 stages is 1:1:3:1, and the number of blocks is L. L is set to 6, 12, 18, 24 respectively, and four derived networks are obtained. These four derived networks include four stages. MetaFormer's token mixer is set to Convolutions. 3D model is projected into 24 views, and Mini Batch K-Means algorithm is used to select 12 representative views. MetaFormer is adopted to extract view features from representative views and representative views are classified, based on which majority voting algorithm is used to determine category of 3D model. Experimental results are shown in Table 4.

From Table 4, we can see that when L increases, accuracy of the proposed network on validation set increases firstly and then decreases, and its FLOPs and Params increase. When L is set to 6, accuracy, FLOPs and Params of the proposed network are the lowest, which are respectively 90.8 %, 6.2G and 22.1M. When L is set to 18, accuracy of the proposed network is the highest and the value is 91.9 %, and its FLOPs and Params are respectively 11.2G and 29.1M. At the same time, we can find that when L is set respectively to 12, 18, 24, accuracy of the proposed network is almost the same. But, with L value increasing, FLOPs and Params of the proposed network increase rapidly. So, L is set to 12 in experiments.

The fourth group of experiments investigate the influence of shape features on the proposed network. 8 experiments are conducted, and 3D model is projected into 24 views. Mini Batch K-Means algorithm is used to select 12 representative views, and their shape descriptors D1, D2, D3 are respectively computed. Experiments 1–4 verify the effect of shape features extracted from shape descriptors D1, D2, D3. Experiment 1 uses CNN to extract D1 shape feature from D1 shape descriptor. Experiment 2 adopts CNN to extract D2 shape feature from D2 shape descriptor. Experiment 3 uses CNN to extract D3 shape feature from D3 shape descriptor. Experiment 4 adopts CNN to extract D1+D2+D3 shape feature from D1, D2, D3 shape descriptor. Representative views are classified, and then category of 3D model is determined by majority voting algorithm based on shape features. Experiments 5–8 verify the effect of the fusion of view features and shape features. MetaFormer is used to extract view features from representative views. Experiment 5 uses CNN to extract D1 shape feature from D1 shape descriptor. Experiment 6 adopts CNN to extract D2 shape feature from D2 shape descriptor. Experiment 7 uses CNN to extract D3 shape feature from D3 shape descriptor. Experiment 8 adopts CNN to extract D1+D2+D3 shape feature from D1, D2, and D3 shape descriptors. View features are fused with shape features extracted from Experiments 5, 6, 7, 8 respectively to

Table 2
The comparison of K-Means and Mini Batch K-Means.

Method	Average iteration time(s)	Peak memory (GB)
K-Means	15.4	3.1
Mini Batch K-Means	5.1	1.5

Experimental results show that compared with K-Means, average iteration time of Mini Batch K-Means reduces from 15.4s to 5.1s, and its memory storage reduces from 3.1 GB to 1.5 GB. We can see that Mini Batch K-Means is more suitable for a large scale of view clustering tasks.

Table 4
Accuracies, FLOPs and Params of the proposed network on validation set under different L value.

L	Accuracy	FLOPs(G)	Params(M)
6	0.908	6.2	22.1
12	0.917	8.7	25.6
18	0.919	11.2	29.1
24	0.916	13.7	32.6

classify representative views. Then, category of 3D model is determined by majority voting algorithm based on the fusion of view features and shape features. Training data is used to optimize these 8 networks, and test data is used to testify these 8 networks. Experimental results are shown in Table 5.

It can be seen from Table 5 that D1 shape feature has the best ability to describe 3D model, and the description ability of D3 shape feature is the worst. The description ability of D1+D2+D3 shape feature is better than that of D1 shape feature, D2 shape feature and D3 shape feature. When view feature is fused with D1 shape feature or D2 shape feature, accuracy of the proposed network is 92.1 %. When view feature is fused with D3 shape feature, accuracy of the proposed network is 91.9 % and decreases 0.1 %. When view feature is fused with D1, D2, D3 shape feature, the proposed network achieves the best and its accuracy is 92.4 %.

The fourth group of experiments show that the description ability of the fusion of view feature and shape feature is stronger than that of shape feature. When view feature is fused with D1+D2+D3 shape feature, 3D model can be described from different perspectives and more comprehensively. 3D model is projected from different angles to get 24 views. Mini Batch K-Means algorithm is adopted to extract representative views from these 24 views. Then, MetaFormer is applied to extract view features from representative views. So, view features can describe local details of 3D model better from different perspectives. CNN is used to extract shape features from shape descriptors D1, D2, D3. So, shape features can express shape and structure of 3D model from different angles, and provide comprehensive information about 3D model. It helps compensate for shortcomings of view feature and effectively improves accuracy of the proposed network.

In order to evaluate the misclassification of the proposed network under each category, data from Experiment 8 in Table 5 is used to construct confusion matrix as shown in Fig. 5.

Accuracies of 7 categories are all above 90 %, including bathtub, bed, chair, monitor, sofa, table and toilet. Here, bathtub category achieves 100 % at accuracy. For desk category, its accuracy is 77 %, 10 % of 3D models are misclassified into table category, and 13 % of models are misclassified into toilet category. Accuracy of dresser category is 86 % and 14 % of 3D models are misclassified into night_stand category. For night_stand category, its accuracy is 86 %, 12 % of 3D models are misclassified into dresser category, and 2 % of models are misclassified into toilet category.

By comparing 3D models, we can find that although local shapes between night_stand and dresser, and local shapes between desk and toilet are different, shapes are relatively similar on the whole. It can easily cause confusion when 3D models are processed and classified.

In order to evaluate the influence of view features extracted by MetaFormer and shape features extracted by CNN on the proposed network, ablation experiments are conducted. 3D model is projected into 24 views. Mini Batch K-Means algorithm is used to select 12 representative views, and their shape descriptors D1, D2, D3 are respectively computed. CNN is used to extract D1+D2+D3 shape feature from D1, D2, D3 shape descriptor. Representative views are classified, and then the category of 3D model is determined by majority voting algorithm based on D1+D2+D3 shape feature. MetaFormer is used to extract view features from representative views. Representative views are classified, and then the category of 3D model is determined by majority voting algorithm based on view features. View features and D1+D2+D3 shape feature are fused to classify representative views. Then, the category of 3D model is determined by majority voting algorithm based on the fusion of view features and D1+D2+D3 shape feature. Training data is used to optimize these 3 networks, and test data is used to testify 3 networks. Experimental results are shown in Table 6.

It can be seen from Table 6 that when view features are only used, the proposed network achieves 88.5 % at accuracy. When D1+D2+D3 shape feature is only adopted, accuracy of the proposed network is 52.8 %. This is because that view features can describe 3D model's global shape information. D1+D2+D3 shape feature can describe 3D model's local contour information, but it can't represent it on the whole. So, the proposed network achieves better classification results based on view features. When view features and D1+D2+D3 shape feature are fused, the proposed network achieves 92.4 % at accuracy. This is because that when view features and D1+D2+D3 shape feature are fused, they can complement each other to better describe global shape and local contour details of 3D model.

The comparison between the proposed method and other view-based ones is shown in Table 7.

From Table 7, we can see that the proposed method achieves the best and its accuracy is 92.40 %. DeepPano has the lowest accuracy, which is 88.66 %. The reason is that view features do not emphasize the contour boundaries of 3D model, and shape features can make up for this shortcoming. When view features and shape features are combined to represent 3D model, we can better capture global and local shape information, which improves accuracy of 3D model classification. Computation costs of these 5 methods are shown in Table 7. Computation complexity of Geometry image (Sinha et al., 2016) is the lowest with 1.9G FLOPs and 4.8M Params. But, its accuracy is the lowest. Computation cost of MVCNN(Su et al., 2015) is the highest, with 15.6G FLOPs and 60.3M Params. The proposed method reduces computation complexity of high-dimensional fusion by clustering representative views, with 8.7G FLOPs and 25.6M Params, which is significantly lower than MVCNN. But, its accuracy is the highest and the value is 92.4 %.

In order to verify the generalization ability of the proposed method on more diverse datasets, experiments are conducted on ModelNet40. The comparison of experimental results on ModelNet10 and ModelNet40 is shown in Table 8.

From Table 8, we can see that the proposed method achieves accuracy of 92.40 % on ModelNet10 and accuracy of 92.70 % on ModelNet40. This shows that the proposed method can accurately determine categories of 3D models in datasets with different sizes and complexities, and achieves higher accuracy. This verifies the proposed method's generalization ability on larger and more diverse datasets.

We compare the proposed network with PointNet++ on ModelNet40

Table 5
Results of 8 experiments.

	Exp 1	Exp 2	Exp 3	Exp 4	Exp 5	Exp 6	Exp 7	Exp 8
bathtub	0.480	0.410	0.370	0.570	1.000	0.990	1.000	1.000
bed	0.450	0.390	0.310	0.500	0.960	0.970	0.960	0.980
chair	0.440	0.320	0.260	0.440	0.970	0.960	0.970	0.970
desk	0.300	0.240	0.190	0.480	0.750	0.770	0.740	0.770
dresser	0.360	0.290	0.220	0.560	0.850	0.850	0.860	0.860
monitor	0.390	0.270	0.230	0.510	0.980	0.980	0.970	0.970
night_stand	0.330	0.280	0.130	0.550	0.860	0.850	0.860	0.860
sofa	0.430	0.320	0.250	0.540	0.980	0.960	0.960	0.970
table	0.380	0.300	0.170	0.540	0.890	0.910	0.910	0.900
toilet	0.420	0.310	0.340	0.590	0.970	0.970	0.960	0.960
Accuracy	0.442	0.313	0.247	0.528	0.921	0.921	0.919	0.924

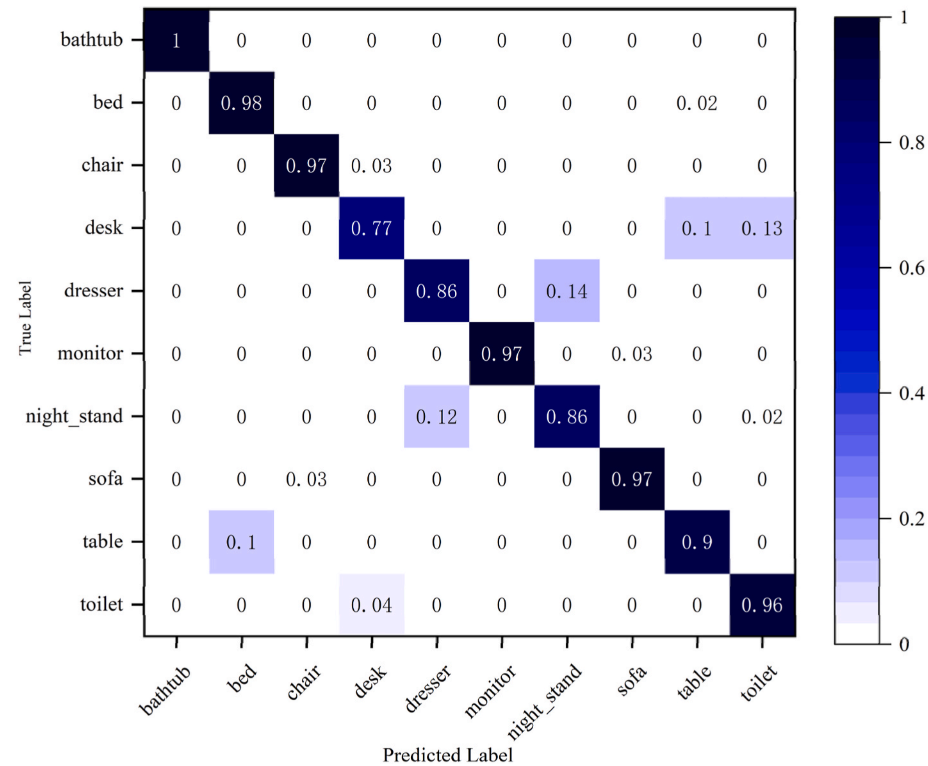

Fig. 5. Confusion matrix of the proposed network.

Table 6
The influence of view features and D1+D2+D3 shape feature on the proposed network on test data.

	Accuracy
View features	0.885
D1+D2+D3 shape feature	0.528
View features + D1+D2+D3 shape feature	0.924

Table 7
The comparison between the proposed method and other view-based ones.

	Number of views	Accuracy	FLOPs (G)	Params (M)
DeepPano (Shi et al., 2015)	1	0.8866	2.1	5.3
PANORAMA-NN(Sfikas et al., 2017)	1	0.9110	3.8	12.7
MVCNN(Su et al., 2015)	20	0.9010	15.6	60.3
Geometry image (Sinha et al., 2016)	1	0.8840	1.9	4.8
The proposed method	12	0.9240	8.7	25.6

dataset. Experimental results of the proposed network and PointNet++ on ModelNet40 are shown in Table 9.

From Table 9, we can see that PointNet++ achieves accuracy of 90.70 % and the proposed network achieves accuracy of 92.70 % on ModelNet40. This shows that the proposed network can determine categories of 3D models better accurately on ModelNet40 compared with PointNet++. The reason is that Mini Batch K-Means algorithm is adopted to extract representative views to reduce the influence of redundant information. View feature is extracted from representative view and shape feature is extracted from shape descriptors of representative view. View feature and shape feature of representative view are fused to describe global shape and local contour details of 3D model more comprehensively. At the same time, majority voting algorithm is used to determine category of 3D model based on the fusion of view features and shape features.

Table 8
Comparison of experimental results on ModelNet10 and ModelNet40.

Datasets	Accuracy
ModelNet10	0.9240
ModelNet40	0.9270

Table 9
Experimental results of the proposed network and PointNet++ on ModelNet40.

Networks	Accuracy
PointNet++(Charles et al., 2017)	0.9070
The proposed network	0.9270

6. Conclusions

As the scale of 3D models continues to increase, the accuracy and efficiency of 3D model classification are facing huge challenges. This paper proposes a method of 3D model classification which fuses MetaFormer and CNN. A fixed-view projection method is used to obtain 2D views of 3D model, and Mini Batch K-Means algorithm is adopted to select 12 representative views. MetaFormer is used to obtain view features from representative views. Points are randomly sampled from representative views, and shape descriptors D1, D2, and D3 are calculated. At the same time, CNN is adopted to obtain shape features from shape descriptors D1, D2, and D3. View features and shape features of representative views are combined to classify 2D views, and majority voting algorithm is used to classify 3D models. Experimental results show that compared with other view-based classification methods, accuracy of the proposed method is improved.

CRediT authorship contribution statement

Xueyao Gao: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Data curation. **Haomin Wu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Data curation. **Chunxiang Zhang:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Yongzeng Xue:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

Bai, J., Si, Q.L., Qin, F.W., 2019. 3D model classification and retrieval based on cnn and voting scheme. J. Computer-Aided Des. Comput. Graph. 31 (2), 303–314. https://doi.org/10.3724/SP.J.1089.2019.17160.

Cai, W.W., Liu, D., Ning, X., et al., 2021. Voxel-based three-view hybrid parallel network for 3d object classification. Displays 69, 1–8. https://doi.org/10.1016/j.displa.2021.102076.

Charles, R.Q., Li, Y., Hao, S., et al., 2017. Pointnet++: deep hierarchical feature learning on point sets in a metric space. In: The 31st Conference on Neural Information Processing Systems, pp. 1–10. https://doi.org/10.48550/arXiv.1706.02413.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al., 2021. An image is worth 16x16 words: transformers for image recognition at scale. In: The 9th International Conference on Learning Representations, pp. 1–8. https://doi.org/10.48550/arXiv.2010.11929.

Gao, X.Y., Wang, Y.Z., Zhang, C.X., et al., 2021. Multi-head self-attention for 3d point cloud classification. IEEE Access 9, 18137–18147. https://doi.org/10.1109/ACCESS.2021.3050488.

Guo, R., Zhou, Y., Zhao, J.Q., et al., 2021. Point cloud classification by dynamic graph cnn with adaptive feature fusion. IET Comput. Vis. 15, 235–244. https://doi.org/10.1049/cvi2.12039.

Guan, R.P., Kuang, L.Q., Jiao, S.C., et al., 2023. Three-dimensional retrieval method driven by multi-modal feature fusion and word embedding. Comput. Eng. 49 (4), 101–107. https://doi.org/10.19678/j.issn.1000-3428.0064951. +113.

Hassan, R., Fraz, M.M., Rajput, A., et al., 2023. Residual learning with annularly convolutional neural networks for classification and segmentation of 3D point clouds. Neurocomputing 526, 96–108. https://doi.org/10.1016/j.neucom.2023.01.026.

Huang, G., Liu, Z., Laurens, V.D.M., et al., 2017. Densely connected convolutional networks. In: The 30th IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 2261–2269. https://doi.org/10.1109/CVPR.2017.243.

Huang, Q., Wang, Y.X., Yin, Z., et al., 2020. View-based weight network for 3d ob ject recognition. Image Vis Comput. 93, 1–16. https://doi.org/10.1016/j.imavis.2019.11.006.

Jin, X., Li, D., 2022. Rotation prediction based representative view locating frame work for 3d object recognition. Comput. Aided Des. 150, 1–11. https://doi.org/10.1016/j.cad.2022.103279.

Kim, S., Chi, H.G., Ramani, K., 2021. Object synthesis by learning part geometry with surface and volumetric representations. Comput. Aided Des. 130, 1–10. https://doi.org/10.1016/j.cad.2020.102932.

Liang, J.A., Chen, K., Ji, Z., et al., 2023. Analysis of learning strategy of deep learn ing classification model for 3d point cloud—taking airborne multispectral lidar point cloud as an example. Surveying Mapping Eng. 32, 68–75. https://doi.org/10.19349/j.cnki.issn1006-7949.2023.06.010.

Liang, Q., Wang, Y.X., Nie, W.Z., et al., 2020. Mvcln: multi-view convolutional lstm network for cross-media 3d shape recognition. IEEE Access 8, 139792–139802. https://doi.org/10.1109/ACCESS.2020.3012692.

Liu, M., Liu, Y., Ren, Q.Q., 2025. Rethinking multi-level information fusion in tem poral graphs: pre-training then distilling for better embedding. Inf. Fusion 121, 1–12. https://doi.org/10.1016/j.inffus.2025.103127.

Ma, Z.P., Zhou, J., Ma, J.L., et al., 2022. A novel 3d shape recognition method based on double-channel attention residual network. Multimed. Tool. Appl. 81 (22), 32519–32548. https://doi.org/10.1007/s11042-022-12041-9.

Osada, R., Funkhouser, T., Chazelle, B., et al., 2001. Matching 3d models with shape distributions. In: International Conference on Shape Modeling and Applications, pp. 154–166. https://doi.org/10.1109/SMA.2001.923386.

Sandler, M., Howard, A., Zhu, M.L., et al., 2018. Mobilenetv2: inverted residuals and linear bottlenecks. In: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 4510–4520. https://doi.org/10.1109/CVPR.2018.00474.

Shi, B., Bai, S., Zhou, Z., et al., 2015. Deeppano: deep panoramic representation for 3d shape recognition. IEEE Signal Process. Lett. 22 (12), 2339–2343. https://doi.org/10.1109/LSP.2015.2480802.

Sfikas, K., Theoharis, T., Pratikakis, I., 2017. Exploiting the panorama representation for convolutional neural network classification and retrieval. In: Eurographics Workshop on 3D Object Retrieval, pp. 1–7. https://doi.org/10.2312/3dor.20171045.

Sinha, A., Bai, J., Ramani, K., 2016. Deep learning 3d shape surfaces using geometry images. In: The 14th European Conference on Computer Vision, pp. 223–240. https://doi.org/10.1007/978-3-319-46466-4_14.

Song, Y.P., He, F.Z., Duan, Y.S., et al., 2022. A kernel correlation-based approach to adaptively acquire local features for learning 3d point clouds. Comput. Aided Des. 146, 1–12. https://doi.org/10.1016/j.cad.2022.103196.

Su, H., Maji, S., Kalogerakis, E., et al., 2015. Multi-view convolutional neural net works for 3d shape recognition. In: 2015 IEEE International Conference on Computer Vision (ICCV), pp. 945–953. https://doi.org/10.1109/ICCV.2015.114.

Szegedy, C., Liu, W., Jia, Y.Q., et al., 2015. Going deeper with convolutions. In: 2015 IEEE Conference on Computer Vision and Pattern Recognition, pp. 1–9. https://doi.org/10.1109/CVPR.2015.7298594.

Tolstikhin, I., Houlsby, N., Kolesnikov, A., et al., 2021. Mlp-mixer: an all-mlp archi tecture for vision. In: The 35th Conference on Neural Information Processing Systems, pp. 24261–24272. https://doi.org/10.48550/arXiv.2105.01601.

Touvron, H., Bojanowski, P., Caron, M., et al., 2023. Resmlp: feedforward networks for image classification with data-efficient training. IEEE Trans. Pattern Anal. Mach. Intell. 45 (4), 5314–5321. https://doi.org/10.48550/arXiv.2105.03404.

Vaswani, A., Shazeer, N., Parmar, N., et al., 2017. Attention is all you need. In: The 31st Annual Conference on Neural Information Processing Systems, vol. 30, pp. 1–8. https://doi.org/10.48550/arXiv.1706.03762.

Yang, J., Wang, S., Zhou, P., 2019. Recognition and classification for three-dimensional model based on deep voxel convolution neural network. Acta Opt. Sin. 39 (4), 1–11. https://doi.org/10.3788/AOS201939.0415007.

Yu, W., Luo, M., Zhou, P., et al., 2022. Metaformer is actually what you need for vi sion. In: 2022 IEEE International Conference on Computer Vision, pp. 10809–10819. https://doi.org/10.48550/arXiv.2111.11418.

Zhang, Z.Z., Lin, H.J., Zhao, X.B., et al., 2018. Inductive multi-hypergraph learning and its application on view-based 3D object classification. IEEE Trans. Image Process. 27 (2), 5957–5968. https://doi.org/10.1109/TIP.2018.2862625.

Zhu, F., Xu, J.Y., Yao, C.M., et al., 2022. Local information fusion network for 3d shape classification and retrieval. Image Vis Comput. 121, 1–7. https://doi.org/10.1016/j.imavis.2022.104405.